

## 2020 ICM Weekend 1

### Problem E: Drowning in Plastic

Since the 1950s, the manufacturing of plastics has grown exponentially because of its variety of uses, such as food packaging, consumer products, medical devices, and construction. While there are significant benefits, the negative implications associated with increased production of plastics are concerning. Plastic products do not readily break down, are difficult to dispose of, and only about 9% of plastics are recycled<sup>[1]</sup>. Effects can be seen by the approximately 4-12 million tons of **plastic waste** that enter the oceans each year<sup>[1,2]</sup>. Plastic waste has severe environmental consequences and it is predicted that if our current trends continue, the oceans will be filled with more plastic than fish by 2050<sup>[2]</sup>. The effect on marine life has been studied<sup>[3]</sup>, **but the effects on human health are not yet completely understood**<sup>[4]</sup>. The rise of **single-use** and **disposable plastic products** results in entire industries dedicated to creating plastic waste. It also suggests that the amount of time the product is useful is significantly shorter than the time it takes to properly **mitigate** the plastic waste. Consequently, to solve the plastic waste problem, we need to slow down the flow of plastic production and improve how we manage plastic waste.

Your team has been hired by the International Council of Plastic Waste Management (ICM) to address this escalating environmental crisis. You must develop a plan to significantly reduce, if not eliminate, single-use and disposable plastic product waste.

- Develop a model **to estimate the maximum levels of single-use or disposable plastic product waste that can safely be mitigated without further environmental damage**. You may need to consider, among many factors, the source of this waste, the extent of the current waste problem, and the availability of resources to process the waste.
- Discuss **to what extent plastic waste can be reduced to reach an environmentally safe level**. This may involve considering factors impacting the levels of plastic waste to include, but not limited to, **sources and uses of single-use or disposable plastics, the availability of alternatives to plastics, the impact on the lives of citizens, or policies of cities, regions, countries, and continents to decrease single-use or disposable plastic and the effectiveness of such policies**. These can vary between regions, so considering **regional-specific constraints** may make some policies more effective than others.
- Using your model and discussion, set a target for the minimal achievable level of global waste of single-use or disposable plastic products and discuss the impacts for achieving such levels. You may consider ways in which human life is altered, the environmental impacts, or the effects on the multi-trillion-dollar plastic industry.
- While this is a global problem, the causes and effects are not equally distributed across nations or regions. Discuss the equity issues that arise from the global crisis and your intended solutions. How do you suggest ICM address these issues?
- Write a two-page memo to the ICM describing a realistic global target minimum achievable level of global single-use or disposable plastic product waste, a timeline to

reach this level, and any circumstances that may accelerate or hinder the achievement of your target and timeline.

Your submission should consist of:

- One-page Summary Sheet
- Table of Contents
- Two-page Memo
- Your solution of no more than 20 pages, for a maximum of 24 pages with your summary, table of contents, and two-page memo.

Note: Reference List and any appendices do not count toward the page limit and should appear after your completed solution. You should not make use of unauthorized images and materials whose use is restricted by copyright laws. Ensure you cite the sources for your ideas and the materials used in your report.

## Glossary

**Disposable Plastic Products:** plastic materials or products that are not recyclable and become trash.

**Mitigate:** To make less severe, to moderate, to alleviate.

**Plastic Waste:** plastic objects that have not been recycled properly or cannot be recycled; debris made of plastic.

**Single-Use Plastic Products:** products made of plastic intended for one time use before being discarded.

## Cited References

[1] Geyer, R., Jambeck, J. R., & Law, K. L. (2017). Production, use, and fate of all plastics ever made. *Science Advances*, 3(7), e1700782.

[2] Jambeck, J. R., Geyer, R., Wilcox, C., Siegler, T. R., Perryman, M., Andrady, A., ... & Law, K. L. (2015). Plastic waste inputs from land into the ocean. *Science*, 347(6223), 768-771.

[3] Li, W. C., Tse, H. F., & Fok, L. (2016). Plastic waste in the marine environment: A review of sources, occurrence and effects. *Science of the Total Environment*, 566, 333-349.

[4] Galloway T.S. (2015) Micro- and Nano-plastics and Human Health. In: Bergmann M., Gutow L., Klages M. (eds) *Marine Anthropogenic Litter*.